

Definition

Tx Goals

Medications

Cases

Hypertensive Emergency

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Cases

Other Comorbidities

1. Eclampsia
2. Thrombotic microangiopathy
3. Pheochromocytoma
4. Ingestion of sympathomimetic agents

CNS

1. Hypertensive encephalopathy
2. IPH or SAH
3. Ischemic CVA

Cardiac

1. Acute LV failure
2. ACS
3. Aortic dissection

Renal

Rapidly $\uparrow$ Cr

Hypertensive Emergency

Confusion, headache, seizures
Focal neurologic deficits
Blurry vision/ papilledema

Chest pain
Dyspnea
Widened mediastinum
Differential BPs, $\downarrow$ LE pulses

$\downarrow$ UOP
Microscopic hematuria

Pregnancy >20 weeks
Episodic HA, sweating, flushing, tachycardia, hyperthermia
+ urine tox

Severe hypertension (BP $> 180/120$)

End organ damage?

+

-

HTN emergency

HTN urgency

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Rapidly ↑ Cr

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Hypertensive Emergency

Severe hypertension (BP > 180/120)

End organ damage?

+

-

HTN emergency

HTN urgency

ICU, IV BP meds

↑ PO BP meds

Special considerations

- IPH/SAH – SBP <140 ASAP
- Aortic dissection – SBP <120 ASAP

General principles

- 1st h – ↓ BP by <25%
- 4-6 h – BP <160/100
- >24-48 h – BP <140/90

Definition

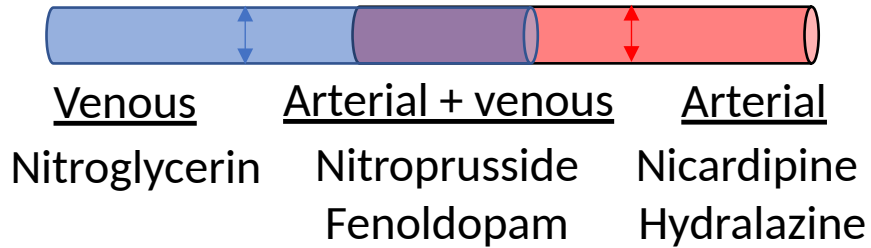
Tx Goals

Medications

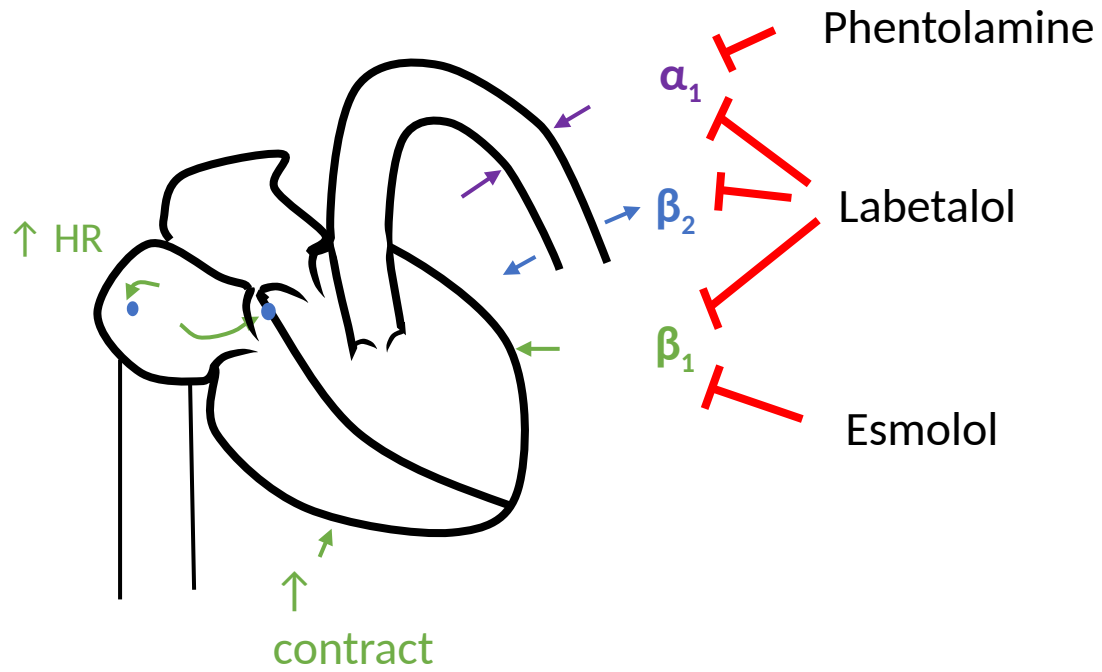
Cases

Hypertensive Emergency

Vasodilators



Adrenergic Antagonists



Severe hypertension (BP > 180/120)

End organ damage?

+

-

HTN emergency

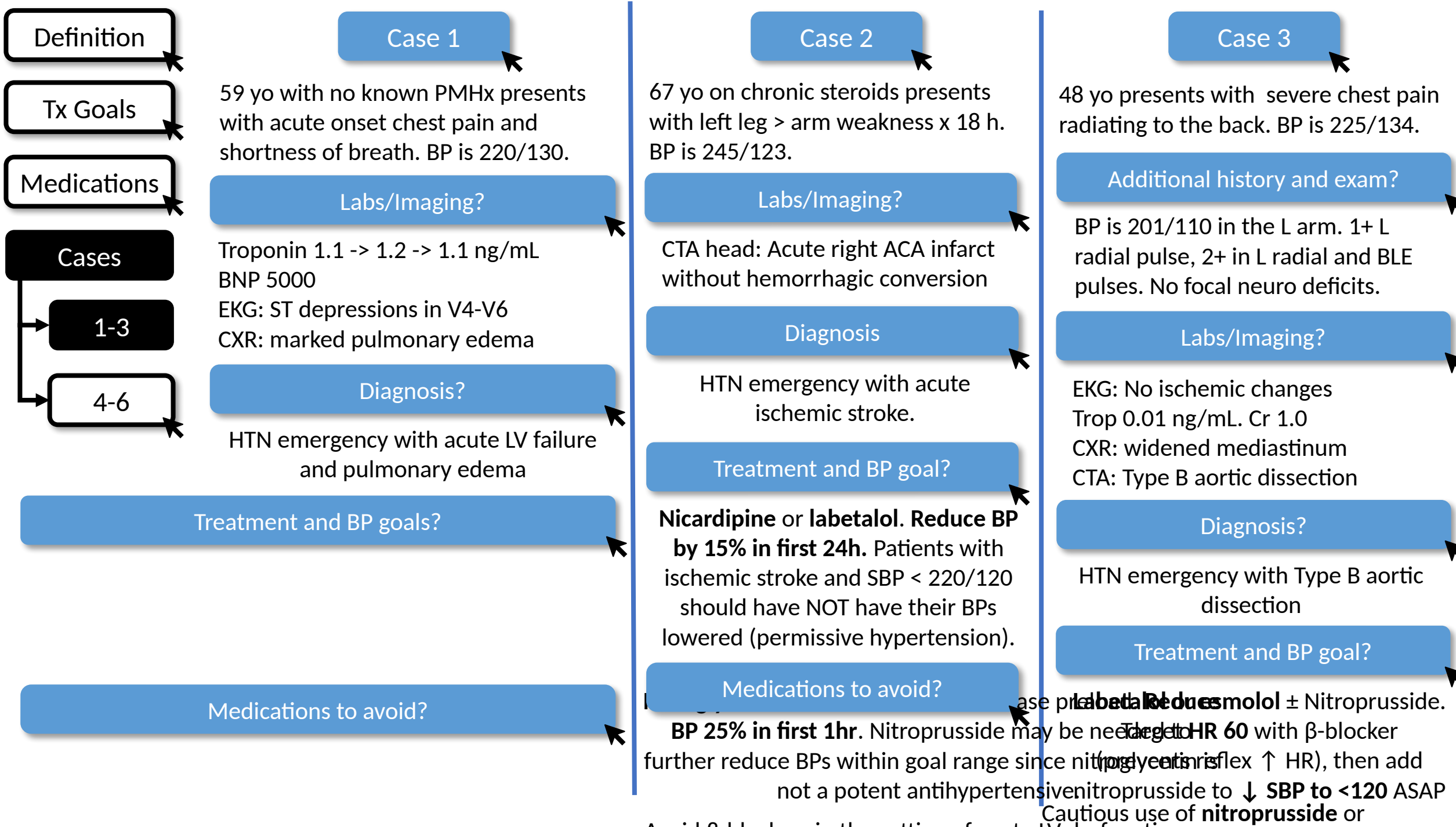
HTN urgency

ICU, IV BP meds

↑ PO BP meds

- IPH/SAH – SBP <140 ASAP
- Aortic dissection – SBP <120 ASAP

- 1st h – ↓ BP by <25%
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- >24 -48 h – BP <140/90



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1-3

4-6

Case 4

67 yo with hypertension presents with nausea/ vomiting x 3 days. BP is 190/125. Home meds include lisinopril 40 mg, amlodipine 10 mg and HCTZ 12.5 mg daily. No meds in the last 3 days.

Additional history and exam?

Reports headache. Denies blurry vision, chest pain, SOB, numbness/ weakness, or confusion. No focal neurologic deficits on exam.

Labs/Imaging

EKG: No ischemic changes
Troponin 0.01 ng/mL. Cr: 1.0
CXR: no focal infiltrates.

Diagnosis?

HTN urgency (or uncontrolled severe HTN).

Treatment?

Control nausea. Restart home meds to ↓ BP over days.

Case 5

47 yo with episodic headaches palpitations, diaphoresis presents with confusion, and fevers. T 40C, BP 243/143, HR 140s.

Lab/Imaging?

EKG: sinus tachycardia
Cr 2.4 (previously normal)
CT A/P: 1.4 cm R adrenal mass
↑ plasma and urine metanephrines

Diagnosis?

Pheochromocytoma crisis

Treatment and BP goal?

Phentolamine, nicardipine or nitroprusside.

↓ SBP to <140 in first 1hr

Medications to avoid?

Case 6

51 yo presents with malaise and decreased urine output. BP is 197/103.

Labs/Imaging?

BMP: K 5.6, HCO₃ 17, Cr 6.7
Renal US: no hydronephrosis or obstruction

Diagnosis?

HTN emergency with acute renal failure

Treatment and BP goal?

2025 Guideline Update — Terminology

- **2025 AHA/ACC HTN guideline retires the term "hypertensive urgency."**
- New label: "severe hypertension WITHOUT acute target-organ damage" (BP >180/120, no end-organ injury). More precise; discourages unnecessary ED visits, IV therapy, and rapid lowering.
- Management of this group is UNCHANGED: reinstitute/intensify ORAL therapy, outpatient with close follow-up. No parenteral drugs, no rapid lowering.
- "Hypertensive emergency" (severe HTN WITH acute end-organ damage) is unchanged: ICU, staged BP lowering, parenteral antihypertensives.
- BP diagnosis/treatment threshold remains 130/80 (2017 reaffirmed); the standing pathophysiology and drug teaching in this talk are still current.

Source: 2025 AHA/ACC Hypertension Guideline (Circulation/Hypertension 2025).

Take-Home Points

- **Hypertensive EMERGENCY = severe HTN (BP >180/120) WITH acute end-organ damage; the end-organ damage — not the BP number — defines it.**
- Screen three organ systems: CNS (encephalopathy, ICH/SAH, ischemic stroke), cardiac (acute LV failure, ACS, aortic dissection), renal (rapidly rising Cr $\pm$ low UOP).
- Goal is to PREVENT further organ damage, not normalize BP: lower SBP $\leq 25\%$ in hour 1 $\rightarrow$ 160/100 by 2–6 h $\rightarrow$ normal over 24–48 h. ICU + IV agents.
- Exceptions need faster, deeper lowering: aortic dissection (HR 60 first with β -blocker, then SBP <120 ASAP), ICH/SAH (SBP <140), eclampsia. Stroke: permissive HTN unless >220/120, or <185/110 before tPA.
- Match drug to scenario: nicardipine (stroke, pregnancy), nitroglycerin (ACS/pulmonary edema), labetalol/esmolol (dissection), phentolamine (catecholamine excess); avoid β -blockers in pheochromocytoma crisis (unopposed α) and in acute LV failure.

Attribution & Optimization Notes

- **Original: "Hypertensive Emergency," TeachIM.org — Sarah Slaven, Christine Sailer, Yilin Zhang MD. Published September 2021.**
- Source URL: teachim.org/teaching_material/htn-emergency/
- License: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Educational use only; not medical advice.
- Corrections (this copy): drug-name spelling "Fenoldapam"/"fendolapam" → "Fenoldopam" (correct INN, dopamine-1 agonist) on slides 4, 6, 7.
- Added context: 2025 AHA/ACC terminology update slide ("hypertensive urgency" → "severe HTN without acute target-organ damage"); standing teaching verified current.
- All original animated slides and graphics preserved; content appended only. Optimized 2026-06-10. Clinical content should be clinician-reviewed before teaching.